

Metoda Gauss

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{pmatrix} \rightarrow \text{matrice echelonați/de tip scara}$$

Exerciții

1) Rezolvați sistemul
$$\begin{cases} 2x_2 + 3x_3 = 8 \\ 2x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 - 2x_3 = -5 \end{cases}$$

$$A = \begin{pmatrix} 0 & 2 & 3 & 8 \\ 2 & 3 & 1 & 5 \\ 1 & -1 & -2 & -5 \end{pmatrix} \xrightarrow{L_1 \leftrightarrow L_3} \begin{pmatrix} 1 & -1 & -2 & -5 \\ 2 & 3 & 1 & 5 \\ 0 & 2 & 3 & 8 \end{pmatrix} \xrightarrow{L_2 - 2L_1} \begin{pmatrix} 1 & -1 & -2 & -5 \\ 0 & 5 & 5 & 15 \\ 0 & 2 & 3 & 8 \end{pmatrix} \xrightarrow{\frac{1}{5}L_2} \begin{pmatrix} 1 & -1 & -2 & -5 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 3 & 8 \end{pmatrix} \xrightarrow{L_3 - 2L_2} \begin{pmatrix} 1 & -1 & -2 & -5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow \text{sistem echivalent: } \begin{cases} x_1 - x_2 - 2x_3 = -5 \\ x_2 + x_3 = 3 \\ x_3 = 2 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 1 \\ x_3 = 2 \end{cases}$$

2)
$$\begin{cases} w - x - y + 2z = 1 \\ 2w - 2x - y + 3z = 3 \\ -w + x - y = -3 \end{cases} \quad A = \begin{pmatrix} 1 & -1 & -1 & 2 & 1 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{pmatrix} \xrightarrow{L_2 - 2L_1, L_3 + L_1} \begin{pmatrix} 1 & -1 & -1 & 2 & 1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & -2 \end{pmatrix} \xrightarrow{L_3 + 2L_2} \begin{pmatrix} 1 & -1 & -1 & 2 & 1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

rang A = nr. linii nenule din forma echelonați

$$\text{rang } A = 2 \quad \text{nr. nec.} = 4 \quad \Rightarrow \text{si. dublu solut.} \Rightarrow \begin{cases} w - x - y + 2z = 1 \\ y - z = 1 \end{cases}$$

Fie $x = \alpha$ cu $\alpha, \beta \in \mathbb{R} \Rightarrow w = \alpha + \beta - 2z$ / $w = \alpha + \beta - 2\beta + 2$
 $y = \beta$ $z = \beta - 1$ $w = \alpha - \beta + 2$

$$(w, x, y, z) \in \{(\alpha - \beta + 2, \alpha, \beta, \beta - 1) \mid \alpha, \beta \in \mathbb{R}\}$$

3)
$$\begin{cases} x_1 - 2x_2 + x_3 + x_4 = 1 \\ x_1 - 2x_2 + x_3 - x_4 = -1 \\ x_1 - 2x_2 + x_3 + 5x_4 = 6 \end{cases} \Rightarrow A = \begin{pmatrix} 1 & -2 & 1 & 1 & 1 \\ 1 & -2 & 1 & -1 & -1 \\ 1 & -2 & 1 & 5 & 6 \end{pmatrix} \xrightarrow{L_2 - L_1, L_3 - L_1} \begin{pmatrix} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 4 & 5 \end{pmatrix} \xrightarrow{L_3 + 2L_2} \begin{pmatrix} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \text{soluție unică}$$

Determinarea inversei unei matrice

$$A^{-1} = ?$$

$$(A; I) \rightarrow (I; A^{-1})$$

4) Determinați A^{-1} pt. care $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 2 & 4 \\ 1 & 3 & -3 \end{pmatrix}$

$$(A; I) = \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 2 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & -3 & 0 & 0 & 1 \end{array} \right) \xrightarrow{L_2 - 2L_1, L_3 - L_1} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 6 & -2 & 1 & 0 \\ 0 & 1 & -2 & -1 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{L_3 + \frac{1}{2}L_2} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 6 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & \frac{1}{2} & 1 \end{array} \right) = \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 6 & -2 & 1 & 0 \\ 0 & 0 & 2 & -4 & 1 & 2 \end{array} \right)$$

$$\xrightarrow{L_2 - 3L_3} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 0 & 10 & -2 & -6 \\ 0 & 0 & 2 & -4 & 1 & 2 \end{array} \right) \xrightarrow{2L_3 + L_2} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 0 & 10 & -2 & -6 \\ 0 & 0 & 2 & -4 & 1 & 2 \end{array} \right)$$

$$\xrightarrow{L_1 + 2L_2} \left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 18 & -3 & -10 \\ 0 & -2 & 0 & 10 & -2 & -6 \\ 0 & 0 & 2 & -4 & 1 & 2 \end{array} \right) \xrightarrow{L_1: 2, L_2: (-2), L_3: 2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 9 & -\frac{3}{2} & -5 \\ 0 & 1 & 0 & -5 & 1 & 3 \\ 0 & 0 & 1 & -2 & \frac{1}{2} & 1 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 9 & -\frac{3}{2} & -5 \\ -5 & 1 & 3 \\ -2 & \frac{1}{2} & 1 \end{pmatrix}$$

5) Determinați A^{-1} pt. $A = \begin{pmatrix} 2 & 1 & -4 \\ -4 & -1 & 6 \\ -2 & 2 & -2 \end{pmatrix}$

$$(A; I) = \left(\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ -4 & -1 & 6 & 0 & 1 & 0 \\ -2 & 2 & -2 & 0 & 0 & 1 \end{array} \right) \xrightarrow{L_2 + 2L_1, L_3 + L_1} \left(\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 2 & 1 & 0 \\ 0 & 3 & -6 & 1 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{L_3 - 3L_2} \left(\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 2 & 1 & 0 \\ 0 & 0 & 0 & -5 & -3 & 1 \end{array} \right) \Rightarrow \text{nu are inversă pt. că}$$

0 linie e nulă

Spatii și subspatii vectoriale

6) Verif. dacă W este subspațiu vectorial al lui V

a) $V = \mathbb{R}^3$ și $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid z = 0 \right\}$

$$\text{I } W \subset V$$

II Fie $w_1, w_2 \in W$ $w_1 = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix}$ $w_2 = \begin{pmatrix} x_2 \\ y_2 \\ 0 \end{pmatrix}$

$$w_1 + w_2 = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ 0 \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \\ 0 \end{pmatrix} \in W$$

III Fie $w_1 \in W$ $w_1 = \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix}$ și $\lambda \in \mathbb{R}$

$$\lambda w_1 = \lambda \begin{pmatrix} x_1 \\ y_1 \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda y_1 \\ 0 \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \\ 0 \end{pmatrix} \in W$$

IV Vectorul nul $0_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \in W$

[Varianta 2] Fie $w_1, w_2 \in W$ și $\alpha, \beta \in \mathbb{R}$

Verif. dacă $\alpha w_1 + \beta w_2 \in W$

b) $V = \mathbb{R}^3$ și $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid z = 1 \right\}$

$$W \subset V \Rightarrow W \text{ e submultime}$$

Vectorul nul $\rightarrow 0_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ pt. că $z = 1$

W nu este subspațiu vectorial al lui V

c) $V = M_n(\mathbb{R})$ $W = \{A \in M_n(\mathbb{R}) \mid A^t = A\}$

$$\text{I } W \subset V$$

II Vector nul $\rightarrow 0_n = 0_n^t$

III Fie $A, B \in W$ a.î. $A^t = A$ și $B^t = B$

$$(A+B)^t = A^t + B^t = A + B = C \in W$$

IV Fie $A \in W$ și $\alpha \in \mathbb{R}$

$$(\alpha A)^t = \alpha \cdot A^t = \alpha \cdot A = C \in W$$

W este subspațiu vectorial al lui V

d) $V = M_2(\mathbb{R})$ $W = \{A \in M_2(\mathbb{R}) \mid A = \begin{pmatrix} a & a+1 \\ 0 & b \end{pmatrix}\}$

$$\text{I } W \subset V$$

II Vector nul $\rightarrow 0_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow a=0, b=0 \Rightarrow a+1=0 \Rightarrow a=-1 \neq 0$

$\Rightarrow W$ nu este subspațiu vectorial al lui V

e) $V = \mathbb{R}_3[x]$ $W = \{P \in \mathbb{R}_3[x] \mid P = a + bx - bx^2 + ax^3\}$

$$\text{I } W \subset V$$

II Vectorul nul $0 = 0 + 0 \cdot x - 0 \cdot x^2 + 0 \cdot x^3 \in W$

III Fie $P_1, P_2 \in W$ a.î. $P_1 = a_1 + b_1x - b_1x^2 + a_1x^3$

$$P_2 = a_2 + b_2x - b_2x^2 + a_2x^3$$

$$P_1 + P_2 = (a_1 + a_2) + (b_1 + b_2)x - (b_1 + b_2)x^2 + (a_1 + a_2)x^3 = a_3 + b_3x - b_3x^2 + a_3x^3 = P_3 \in W$$

IV Fie $P_1 \in W$ și $\alpha \in \mathbb{R} \Rightarrow \alpha P_1 = \alpha(a_1 + b_1x - b_1x^2 + a_1x^3)$

$$\Rightarrow \alpha P_1 = \alpha a_1 + \alpha b_1x - \alpha b_1x^2 + \alpha a_1x^3 = a_2 + b_2x - b_2x^2 + a_2x^3 = P_2 \in W$$

f) $V = \text{spatiul f. continue } f: [a, b] \rightarrow \mathbb{R}$

$$W = \{f \in V \mid f(a) = f(b)\}$$

$$\text{I } W \subset V$$

II Fie $f_1, f_2 \in W$ a.î. $f_1(a) = f_1(b)$ și $f_2(a) = f_2(b)$

$$(f_1 + f_2)(a) \stackrel{f. \text{ cont.}}{=} f_1(a) + f_2(a) = f_1(b) + f_2(b) = (f_1 + f_2)(b) \in W$$

III Fie $\alpha \in \mathbb{R}$ și $f_1 \in W$ cu $f_1(a) = f_1(b)$

$$(cf_1)(a) = c \cdot f_1(a) = c \cdot f_1(b) = (cf_1)(b) \in W$$

IV $f(w) = 0$ a.î. $f(w) + f(a) = f(a) + f(w) = f(a) \Rightarrow 0 \in W$

$\Rightarrow W$ subspațiu vectorial al lui V

7) $A = \begin{pmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$$\text{Ker}(A) = ?$$

$$\text{Ker}(A) = \{v \in \mathbb{R}^3 \mid Av = 0_{\mathbb{R}^3}\}$$

$$\det A = -6 - 12 \cdot 6 + 6 \cdot 8 + 30 = -78 + 78 = 0 \Rightarrow \text{S.C.N.}$$

$$\Delta_p = \begin{vmatrix} 3 & -5 \\ 6 & -2 \end{vmatrix} = -6 + 30 = 24 \neq 0$$

$$\Delta_c = \begin{vmatrix} 3 & -5 & 0 \\ 6 & -2 & 0 \\ -8 & 4 & 0 \end{vmatrix} = 0 \Rightarrow \text{S.C.N.}$$

$$\text{Nec. prin } x, y \Rightarrow \begin{cases} 3x - 5y = 3\alpha \\ 6x - 2y = 0 \end{cases} \Rightarrow y = 3x$$

$$\Rightarrow 3x - 15x = 3\alpha \Rightarrow -12x = 3\alpha \Rightarrow x = -\frac{\alpha}{4}$$

$$y = -\frac{3}{4}\alpha$$

$$(x, y, z) \in \left\{ \left(-\frac{\alpha}{4}, -\frac{3}{4}\alpha, \alpha \right) \mid \alpha \in \mathbb{R} \right\}$$

$$\text{Ker}(A) = \left\{ \begin{pmatrix} -\frac{\alpha}{4} \\ -\frac{3}{4}\alpha \\ \alpha \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$$